

ResQ – Complete Project Overview & Core Features

Extended Core Features and Rationale

1. Proximity-Based Emergency Alerts

When a user triggers an emergency alert (e.g. stealing, harassment, medical distress), the system automatically notifies nearby verified users within a defined radius. This is critical because help from people close by is usually faster than centralized response. **Reason:** Reduces response time and increases chances of immediate intervention, especially for crimes like theft or assault.

2. Emergency Contact Notification

Users can register trusted emergency contacts (family, friends, guardians). When an alert is triggered, these contacts receive instant notifications with location and incident type. **Reason:** Keeps trusted individuals informed and involved, providing emotional support and escalation beyond the app itself.

3. Verified Peer Assistance (Critical Backup)

If no admin or official responder is available, the system allows nearby verified users (e.g. trained volunteers or verified females for sensitive cases) to assist. **Reason:** Ensures no alert goes unanswered, especially in worst-case or highly sensitive emergencies where immediate human presence matters most.

4. Role-Based Access (Resident vs Admin)

Residents can report emergencies and receive alerts. Admins monitor incidents, validate alerts, and coordinate responses. **Reason:** Prevents misuse and ensures accountability.

5. Live Map of Active Emergencies

Displays real-time incidents on a map with severity indicators. **Reason:** Improves situational awareness and enables faster, coordinated responses.

6. Incident History & Reporting

Users can view past incidents they reported. Admins can analyze trends. **Reason:** Supports data-driven safety improvements and academic analysis.

7. Safety Verification & Trust System

Users assisting others must be verified (ID, admin approval, or institutional records). **Reason:** Prevents exploitation and builds trust in peer-to-peer assistance.

8. Fail-Safe Emergency Escalation

If alerts fail locally, the system escalates to admins and emergency contacts. **Reason:** Guarantees layered safety coverage.